DO NOW — RECALL

Tree traversal: post-order and breadth-first

Name: _____ Class: _____ Date: _____

1. What does a constructor do?

2. What is in-order traversal order?

3. What does FIFO stand for?

4. A root R has only left child L. What is post-order?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Initialises an object when it is created.

2. Model answer

Left, root, right.

3. Model answer

First In, First Out.

4. Model answer

L then R.

Vocabulary — know these terms

post-order — Traverse left, right, then root.

breadth-first — Visit vertices level by level.

FIFO queue — First-in-first-out structure used by BFS.

leaf — Node without children.

traversal — Systematically visit each node in a structure.

NEW LEARNING

Read, annotate and explain

Depth-first post-order

Visit an entire left subtree, then right subtree, and finally the current node.

Breadth-first level order

Start with the root in a queue. Dequeue one node, visit it and enqueue each of its children.

Compare output

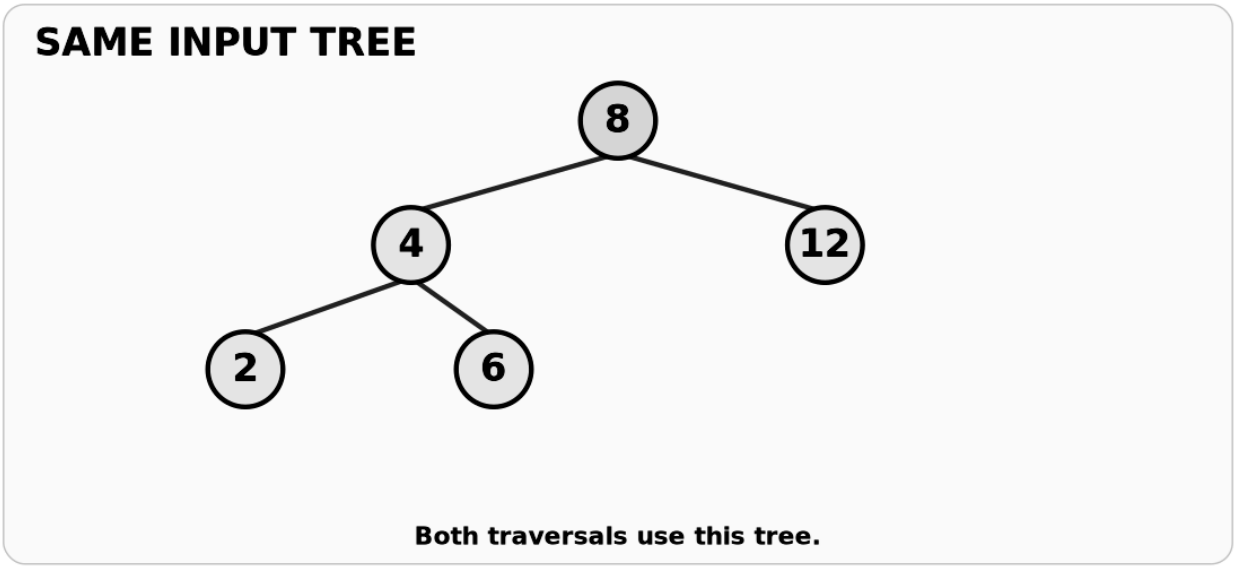
Same structure yields different orders: 2,6,4,12,8 versus 8,4,12,2,6.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

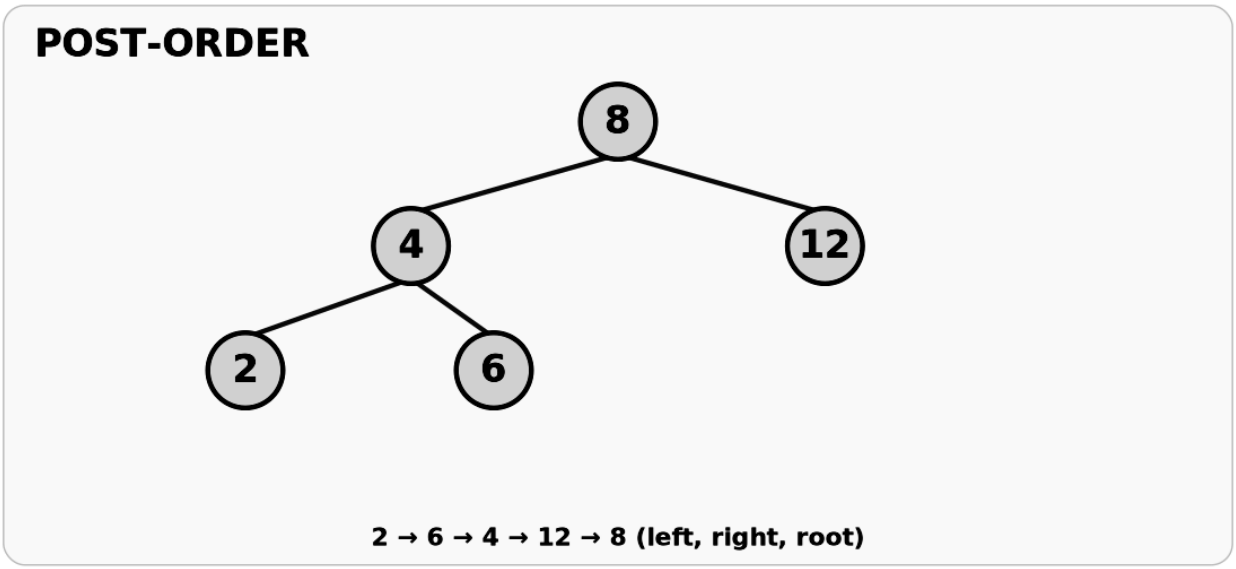
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SAME INPUT TREE: 8, 4, 12, 2, 6. Both traversals use this tree.

New Learning

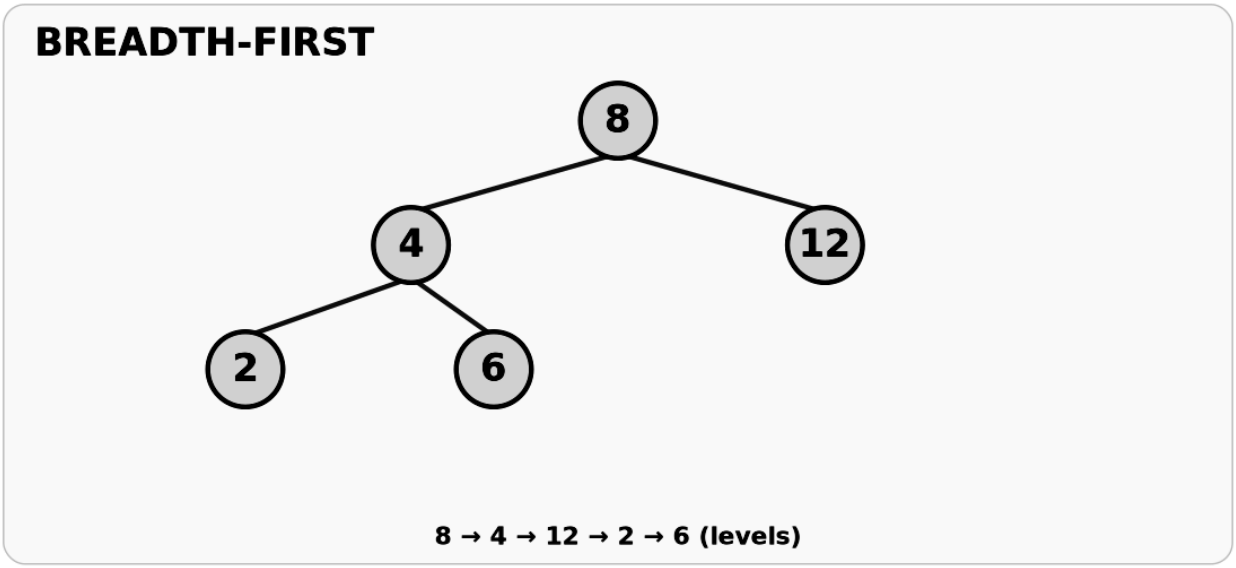


POST-ORDER: 8, 4, 12, 2, 6. 2 → 6 → 4 → 12 → 8 (left, right, root)

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BREADTH-FIRST: 8, 4, 12, 2, 6. 8 → 4 → 12 → 2 → 6 (levels)

I DO — WORKED EXAMPLE

Follow the worked method

Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

Predict first

PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.

Model the reasoning

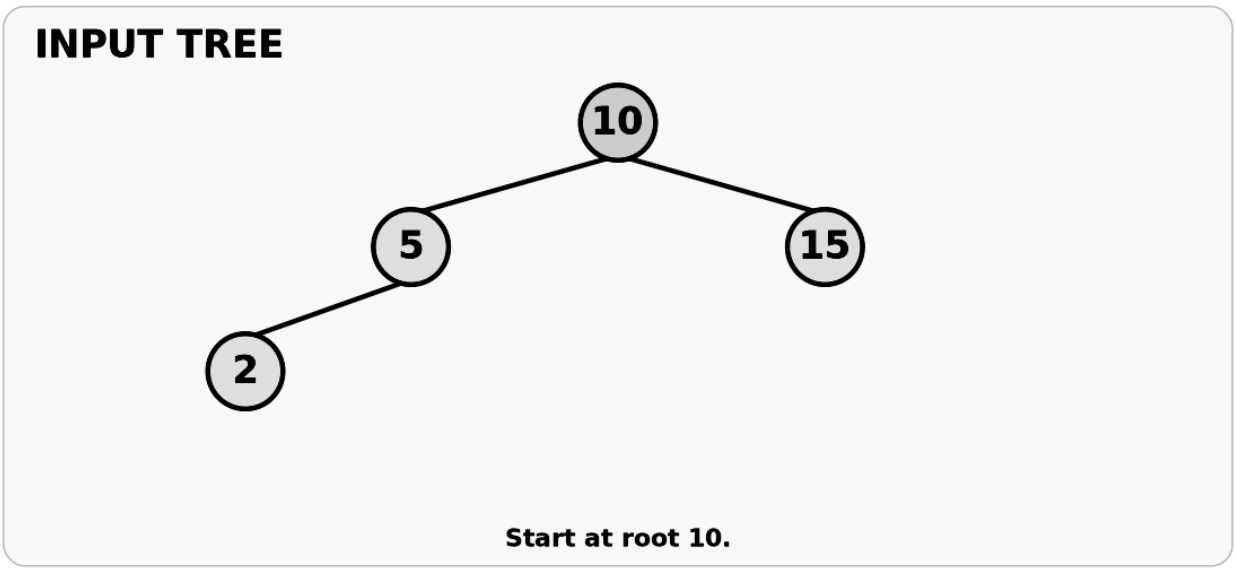
Return to10, traverse right child15 and print15, then print10. Output 2,5,15,10: the print follows both child calls. Node has Key, Left and Right fields.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.



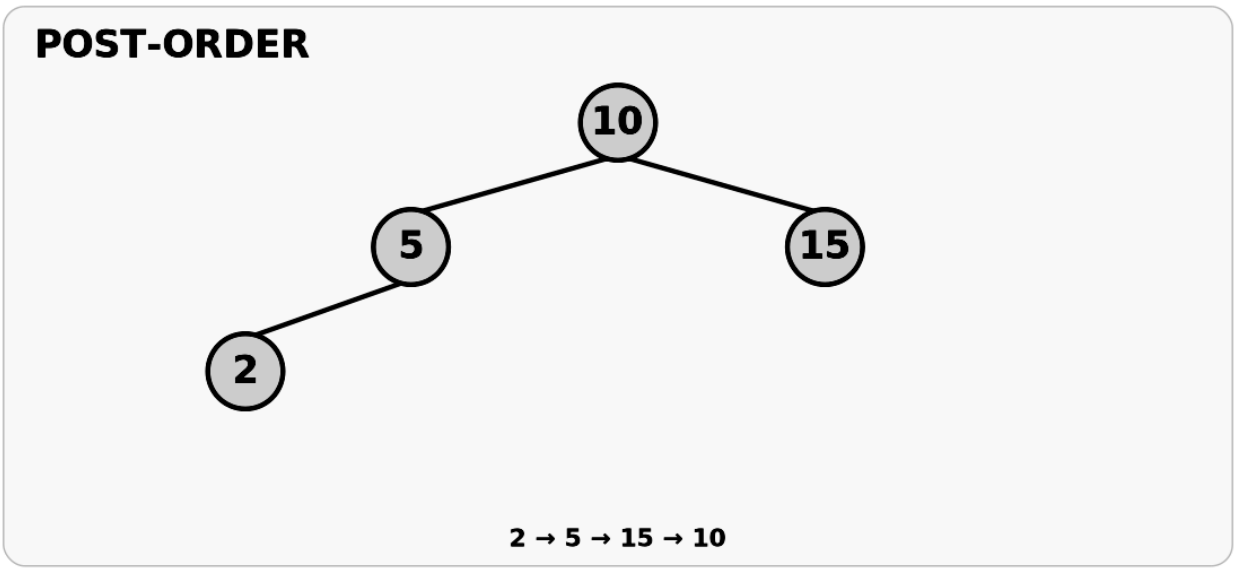
INPUT TREE: 10, 5, 15, 2. Start at root 10.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find post-order traversal of root 10, children 5 and 15, with 5 having left child 2.

PostOrder(10) calls left(5), then left(2). At node2 both null-child calls return, so print2; return to node5 and print5.



POST-ORDER: 10, 5, 15, 2. 2 → 5 → 15 → 10

WE DO — GUIDED PRACTICE

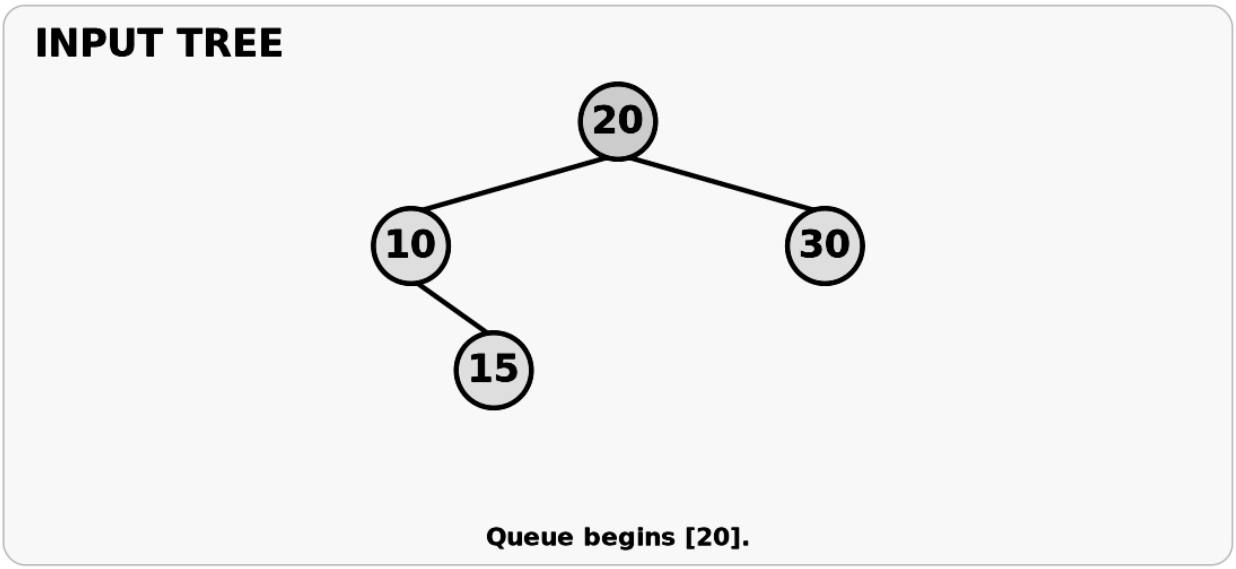
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find breadth-first traversal of root 20, children 10 and 30, with 10 having right child 15.

Visit root 20, enqueue 10 and 30.



INPUT TREE: 20, 10, 30, 15. Queue begins [20].

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Visit root 20, enqueue 10 and 30.

DISCUSS — Compare the sequence of steps or states. Identify the first difference in your methods and justify which step is correct.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A preorder routine prints node.Key, then traverses Left and Right (with a null guard). Move one statement to make it post-order. Give the result for root6 with children3 and9.

Question 2

Tree root 12, left 5 with children 2 and 8, right 17. Give breadth-first and post-order traversals.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write pseudocode to traverse a binary tree breadth-first. Explain why a queue rather than a stack is used.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Breadth-first traversal normally uses a stack to ensure earliest-discovered nodes are visited first.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

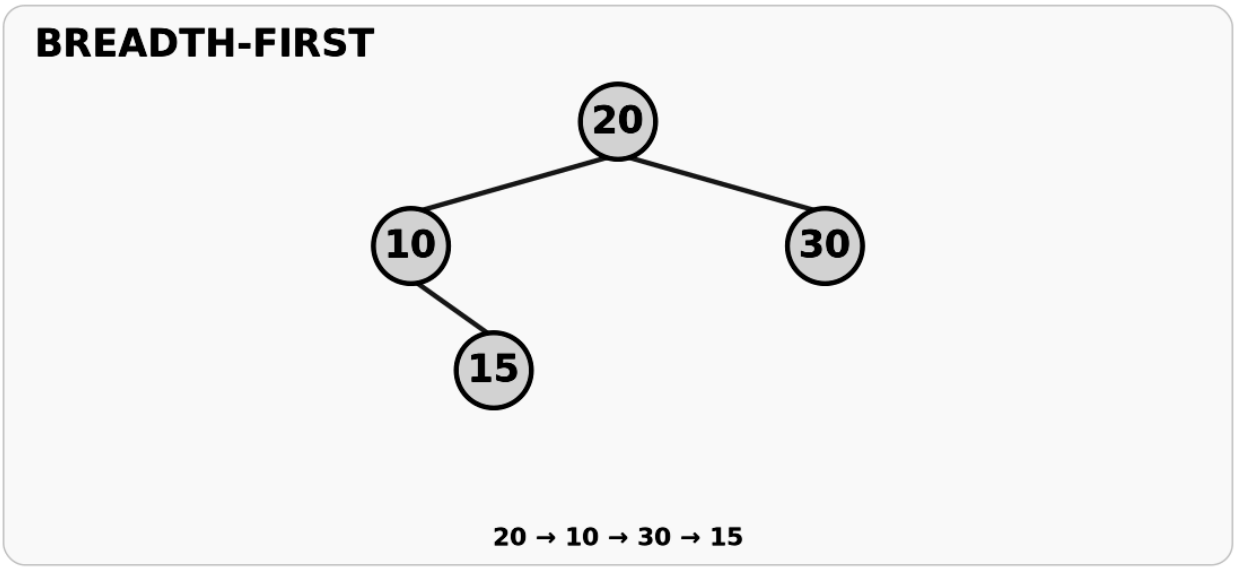
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find breadth-first traversal of root 20, children 10 and 30, with 10 having right child 15.

Visit root 20, enqueue 10 and 30.



BREADTH-FIRST: 20, 10, 30, 15. 20 → 10 → 30 → 15

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model the reasoning

Visit 10 and enqueue child 15, then visit 30; final 15 gives 20,10,30,15.

IMPROVE: Explain one correction to your own reasoning.